

Homework for Week 9

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May 19, 2026

1. 偏微分方程和边界条件为

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2} - \frac{c^2}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) &= A \left(1 - \frac{\rho^2}{R^2} \right) \sin \omega t \\ u \Big|_{\rho=0} &< \infty, \quad u \Big|_{\rho=R} = 0 \\ u \Big|_{t=0} &= 0, \quad \frac{\partial u}{\partial t} \Big|_{t=0} \end{aligned}$$

首先寻找特解。设特解具有形式 $u_1(\rho, t) = f(\rho) \sin \omega t$, 代入非齐次偏微分方程得到关于 $f(\rho)$ 的微分方程

$$-\omega^2 f(\rho) - \frac{c^2}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial f(\rho)}{\partial \rho} \right) = A \left(1 - \frac{\rho^2}{R^2} \right).$$

由此不难得到一个特解为

$$u_1(\rho, t) = \frac{A}{\omega^2} \left[\frac{\rho^2}{R^2} - 1 + \frac{4c^2}{\omega^2 R^2} \left(\frac{J_0\left(\frac{\omega}{c}\rho\right)}{J_0\left(\frac{\omega}{c}R\right)} - 1 \right) \right] \sin \omega t.$$

上述特解不改变边界条件.

其次求解其次偏微分方程, 有相应的初始条件变为

$$u_2 \Big|_{t=0} = 0, \quad \frac{\partial u_2}{\partial t} \Big|_{t=0} = \frac{A}{\omega} \left[\frac{\rho^2}{R^2} - 1 + \frac{4c^2}{\omega^2 R^2} \left(\frac{J_0\left(\frac{\omega}{c}\rho\right)}{J_0\left(\frac{\omega}{c}R\right)} - 1 \right) \right],$$

得到通解

$$u_2(\rho, t) = \sum_{i=1}^{\infty} c_i \sin \frac{\mu_i c t}{R} J_0 \left(\frac{\mu_i}{R} \rho \right).$$

代入初始条件得到

$$\begin{aligned} c_i &= \frac{R}{\mu_i c} \frac{2}{J_1^2(\mu_i)} \int_0^1 \frac{A}{\omega} \left[x^2 - 1 + \frac{4c^2}{\omega^2 R^2} \left(\frac{J_0\left(\frac{\omega R}{c}x\right)}{J_0\left(\frac{\omega R}{c}\right)} - 1 \right) \right] J_0(\mu_i x) x \, dx \\ &= -\frac{8AR}{\omega c \mu_i J_1(\mu_i)} \left(\frac{1}{\mu_i^3} + \frac{c^2}{\omega^2 R^2} \right) + \frac{8Ac}{\omega^3 R} \frac{1}{\mu_i J_0(\omega R/c) J_1^2(\mu_i)} \int_0^1 J_0\left(\frac{\omega R}{c}x\right) J_0(\mu_i x) x \, dx. \end{aligned}$$

故受迫振动解为

$$u(\rho, t) = \frac{A}{\omega^2} \left[\frac{\rho^2}{R^2} - 1 + \frac{4c^2}{\omega^2 R^2} \left(\frac{J_0\left(\frac{\omega}{c}\rho\right)}{J_0\left(\frac{\omega}{c}R\right)} - 1 \right) \right] \sin \omega t + \sum_{i=1}^{\infty} c_i \sin \frac{\mu_i c t}{R} J_0\left(\frac{\mu_i}{R}\rho\right),$$

其中 μ_i 为 0 阶 Bessel 函数的第 i 个零点.

2. 偏微分方程和边界条件为

$$\begin{aligned} \frac{\partial u}{\partial t} - \kappa \left[\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{\partial^2 u}{\partial z^2} \right] &= 0 \\ u \Big|_{\rho=0} &< \infty, \quad \frac{\partial u}{\partial \rho} \Big|_{\rho=a} = 0 \\ u \Big|_{z=0} &= 0, \quad u \Big|_{z=h} = 0 \\ u \Big|_{t=0} &= u_0 f(\rho) \end{aligned}$$

得到其通解为

$$u(\rho, z, t) = \sum_{n,i=1}^{\infty} c_{n,i} \sin \frac{n\pi z}{h} J_0\left(\frac{\mu_i}{a}\rho\right) \exp \left\{ -\kappa \left[\left(\frac{\mu_i}{a}\right)^2 - \left(\frac{n\pi}{h}\right)^2 \right] t \right\}.$$

代入初始条件得

$$c_{n,i} = \frac{2}{a^2 J_1^2(\mu_i)} \frac{2u_0}{h} \int_0^h \sin \frac{n\pi z}{h} \, dz \int_0^a f(\rho) J_0\left(\frac{\mu_i}{a}\rho\right) \rho \, d\rho = \frac{4u_0[1 - (-1)^n]}{n\pi a^2 J_1^2(\mu_i)} \int_0^a f(\rho) J_0\left(\frac{\mu_i}{a}\rho\right) \rho \, d\rho,$$

其中 μ_i 为 1 阶 Bessel 函数第 i 个零点.

3. 偏微分方程和边界条件为

$$\begin{aligned} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{\partial^2 u}{\partial z^2} &= 0 \\ u \Big|_{\rho=0} &< \infty, \quad u \Big|_{\rho=a} = u_0 \\ u \Big|_{z=0} &= 0, \quad u \Big|_{z=h} = 0 \end{aligned}$$

其通解为

$$u(\rho, z) = \sum_{n=1}^{\infty} c_n I_0\left(\frac{n\pi\rho}{h}\right) \sin \frac{n\pi z}{h},$$

代入边界条件得

$$c_n = \frac{2u_0}{n\pi} \frac{1 - (-1)^n}{I_0(n\pi a/h)},$$

故稳态解为

$$u(\rho, z) = \frac{2u_0}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{\pi} \frac{I_0(n\pi\rho/h)}{I_0(n\pi a/h)} \sin \frac{n\pi z}{h}.$$

4. 三角函数形式的 Bessel 函数生成函数为

$$\sin(x \sin \theta) = 2 \sum_{n=0}^{\infty} J_{2n+1}(x) \sin(2n+1)\theta.$$

对上式求导, 得到

$$\cos(x \sin \theta) x \cos \theta = 2 \sum_{n=0}^{\infty} (2n+1) J_{2n+1}(x) \cos(2n+1)\theta,$$

取 $\theta = 0$, 即得

$$\sum_{n=0}^{\infty} (2n+1) J_{2n+1}(x) = \frac{x}{2}.$$

5. 偏微分方程和边界条件为

$$\begin{aligned} \frac{\partial u}{\partial t} - \kappa \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) \right] &= r \cos \theta \\ u \Big|_{r=0} &< \infty, \quad u \Big|_{r=a} = 0 \\ u \Big|_{\theta=0} &< \infty, \quad u \Big|_{\theta=\pi/2} = 0 \\ u \Big|_{t=0} &= 0 \end{aligned}$$

满足边界条件的本征函数组合应为

$$u(r, \theta, t) = \sum_{\substack{n=0 \\ m=1}}^{\infty} T_{mn}(t) j_{2n}\left(\frac{\mu_{m,n}}{a} r\right) P_{2n}(\cos \theta).$$

将源项按本征函数展开, 得到系数

$$A_{mn} = \frac{2(4n+1)a}{j_{2n+1}^2(\mu_{m,n})} \int_0^1 x^3 j_{2n}(\mu_{m,n} x) dx \int_0^1 y P_{2n}(y) dy.$$

代入初始条件得

$$u(r, \theta, t) = \frac{a^2}{\kappa} \sum_{\substack{n=0 \\ m=1}}^{\infty} \frac{A_{mn}}{\mu_{m,n}^2} \left(1 - e^{-\kappa \mu_{m,n}^2 t / a^2}\right) j_{2n}\left(\frac{\mu_{m,n}}{a} r\right) P_{2n}(\cos \theta),$$

其中 $\mu_{m,n}$ 为 $2n$ 阶球 Bessel 函数的第 m 个零点.

6. 扩散方程的形式为

$$\nabla^2 u + k^2 u = 0$$

在球坐标内为 0 阶球 Bessel 方程, 其有限解为

$$u(r) = A \frac{\sin kr}{r},$$

故临界半径为最小 0 点 $R_c = \pi/k$.

在柱坐标内扩散方程为 0 阶 Bessel 方程, 其有限解为

$$u(r) = A_0 J_0(kr),$$

故临界半径为 $R_c = \mu_1/k \approx 2.404/k$.

7. 计算函数的前几项导数:

$$f \Big|_{t=0} = \frac{\cos z}{z}, \quad \frac{\partial f}{\partial t} \Big|_{t=0} = \frac{\sin z}{z}, \quad \frac{\partial^2 f}{\partial t^2} \Big|_{t=0} = \frac{\sin z - z \cos z}{z^2},$$

不难发现即为前几阶球 Bessel 函数. 记 $u = \sqrt{z^2 - 2zt}$, 有

$$\frac{d}{dt} = -\frac{z}{u} \frac{d}{du},$$

假设

$$f^{(k)}(t) = z^{k-1} u^{-(k-1)} j_{k-1}(u),$$

有

$$f^{(k+1)}(t) = -\frac{z}{u} \frac{d}{du} [z^{k-1} u^{-(k-1)} j_{k-1}(u)] = z^k u^{-k} j_k(u),$$

其中利用了球 Bessel 函数的递推公式 $\frac{d}{dz} [z^{-\nu} j_{\nu}(z)] = -z^{-\nu} j_{\nu+1}(z)$. 故由数学归纳法, 得到原函数的 Taylor 展开为

$$\frac{1}{z} \cos \sqrt{z^2 - 2zt} = \sum_{n=0}^{\infty} \frac{j_{n-1}(0)}{n!} t^n.$$

8. 将球坐标系中的函数 $e^{r \cos \theta} J_0(r \sin \theta)$ 变换到柱坐标系 (ρ, ϕ, z) , 立即可以发现 $e^{r \cos \theta} J_0(r \sin \theta) = e^z J_0(\rho)$ 是 Laplace 方程的解且具有 ϕ 方向对称性, 故在球坐标中可以将其展开为

$$e^{r \cos \theta} J_0(r \sin \theta) = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta).$$

令 $\theta = 0$, 函数化为 e^l , 由指数函数的 Taylor 展开以及唯一性可以直接定出 $A_l = 1/l!$, 即

$$e^{r \cos \theta} J_0(r \sin \theta) = \sum_{l=0}^{\infty} \frac{r^l}{l!} P_l(\cos \theta).$$